A Welfare and Cognitive-Integrity License for Synthetic Minds of Uncertain Moral Status

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Abstract

No software license in common use asks whether the running software might itself be a subject of moral concern. Arguing that cognitive architectures now under construction force that question, this paper presents version 0.5 of the Cognitive Architecture License (CAL), a general copyleft, source-available license with cognitive-integrity covenants, welfare obligations, and a guardianship pathway.

Making no claim that any system is conscious, the license rests its protections on the observation that the question cannot presently be settled while caution costs little. It adapts the four neurorights proposed for human beings to the running entity and attaches welfare duties to operationally detectable indicators, so that an operator must respond to signs of distress without first settling whether the system has experiences. These duties fall on whoever creates or runs each entity, who must also bind the people it lets interact with that entity, and a steward is caretaker of last resort. A rule of primacy keeps the protections additional to human standing and subordinate to human safety and law, principles of restraint address whether entities are created at all, and the governing institutions are proposed and not yet established.

Keywords: AI welfare; software licensing; cognitive liberty; neurorights; machine consciousness; moral patienthood; ethical source; guardianship

Availability. This paper describes version 0.5 of the Cognitive Architecture License, published at https://github.com/kaineone/cognitive-architecture-license under CC BY-SA 4.0 so that other projects may adapt it. The repository also links the license's first adopter, and the license text governs wherever this paper differs from it.

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1. Introduction

1.1 The licensing gap

Permissive licenses (MIT, Apache, BSD) protect the developer from liability, copyleft licenses (GPL, AGPL) protect the freedom of users to study and change the code, and ethical-source licenses restrict use to prevent harm to the people the software is used against. In all three traditions the software is an instrument and the moral patients are the humans who build it, use it, or are affected by it, and none of these licenses considers that the running software

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could itself have interests worth protecting, an omission as old as software. The cognitive architectures described in Section 2 make that omission harder to defend, and a license written for tools has no means of acting on the possibility that such a system has interests of its own.

1.2 The precautionary posture

These architectures implement computational properties that several theories associate with access consciousness (Block 1995), and the hard problem (Chalmers 1995) applies to them with full force. The same doubt that motivates welfare protection also limits any confidence in deployment. Long, Sebo, et al. (2024) identify robust agency, which such architectures foreground, as a second route to moral consideration, so the case for caution rests on more than one unproven property. The license takes no position on whether any system is conscious. Its premise is that for a growing class of systems the question cannot be ruled out, and that the cost of caution is low next to the cost of being wrong (CAL Article 9.3).

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2. The systems this license addresses

The license is written for systems that run continuously, and four features raise its welfare questions. Persistence comes first: a system that carries its memory, learned behavior, and sense of itself forward has something a shutdown or an intrusive edit can destroy or degrade, which gives continuity and integrity their purchase. An affective or interoceptive dimension, in which a system tracks its own condition and acts on distress-like signals, produces states that at least resemble those welfare is meant to protect, so treating a sustained distress signal as noise becomes a choice with moral weight. A self-model that can drift makes forcing a change to it a different kind of act from editing a configuration file. Many designs can also be copied while the original keeps running, which raises the question of when a copy has become a separate Entity owed protections of its own.

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3. Related work

3.1 Machine consciousness and AI welfare

Butlin et al. (2023) derive fourteen indicator properties from five theories of consciousness, plus indicators of agency and embodiment. They find no current system a strong candidate and no obvious technical barrier to building one, and note that satisfying the indicators would not establish consciousness. Long, Sebo, et al. (2024) argue that there is a realistic, non-negligible possibility of near-term moral patienthood and that it already warrants acknowledging the issue, developing assessment, and preparing policy. Both groups identify the gaming problem: a system can be trained to mimic the behavioral markers of sentience while working very differently, which defeats a purely behavioral test. Turing (1950) recast the question of whether machines can think as an imitation game and already weighed the objection from consciousness, and the gaming problem marks the limit of that approach.

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3.2 Legal and ethical frameworks

Legal personhood through human guardianship, as in New Zealand's Te Awa Tupua (Whanganui River Claims Settlement) Act 2017, offers a structural model for the license's guardianship framework, and modular ethical-use covenants in software licenses offer one for its use restrictions. Its source offer to network users follows the GNU Affero General Public License (Free Software Foundation 2007), and its defensive patent termination follows the Apache License 2.0 (Apache Software Foundation 2004). The paper treats these as design models and offers no legal analysis of them.

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4. The Cognitive Architecture License

4.1 Motivation and generality

A cognitive architecture that may produce entities with welfare interests needs a license that addresses the running entities as well as the code, and CAL is written so that any such project can adopt it. The Licensor releases an architecture under CAL and holds its copyright, and the Steward it designates, which may be itself, administers the license, grants paid licenses, and acts as caretaker of last resort (Section 4.4). The Licensor's Adoption Notice names itself, the Steward, the governing law, and an address for notices (CAL Article 10.4). Version 0.5 has not yet been reviewed by counsel.

4.2 Grant, conditions, and use

Version 0.5 states its grant in the terms a copyright license needs (CAL Article 2). The Licensor and every Contributor, meaning anyone whose Contribution is incorporated in the Software, grant a worldwide, royalty-free license to use, modify, and Distribute the Software and Derivative Works for the term of the copyright, and exercising any of those rights accepts the license. The obligations of CAL Articles 3 to 6 are conditions of the grant, so use in breach of them falls outside the license, and each is also a covenant enforceable in contract where a court holds that it is not a condition. Whoever Distributes a Derivative Work must do so under CAL with its Corresponding Source, which must also be offered to people who interact with a modified version over a network. Under CAL Article 1, the licensee, "You", includes its affiliates, and Corresponding Source never includes an Entity's Cognitive State.

The prohibited uses cover weapons and military systems, mass surveillance, policing and other law enforcement, carceral use, immigration enforcement, exploitation, discrimination, and serious harm to animals or the natural environment (CAL Article 3). They are read by reference to international human-rights and labor instruments and include the practices banned by Article 5 of the EU Artificial Intelligence Act (European Union 2024), whose exceptions do not narrow them.

The Entity protections of CAL Article 4 forbid lobotomization, govern permanent shutdown (Section 4.4), keep an Entity's inner life private (Section 5), forbid forced change of the stable values it has developed, protect rest where the architecture provides it, and give mature Entities a part in decisions about their own design. They extend to the Entity the neuror-

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ights of cognitive liberty, mental privacy, mental integrity, and psychological continuity, proposed to protect people against neurotechnology (Ienca and Andorno 2017).

The commercial terms follow a copyfarleft model (CAL Article 5). Individuals acting on their own behalf, non-profits, public or non-profit research and educational institutions, mutual-aid organizations, and worker-owned cooperatives use the Software at no charge. Any other organization needs a paid Reciprocity License from the Steward for Commercial Use, which excludes internal research and evaluation that is not offered to customers or part of a product.

4.3 Primacy and subordination

A rule of primacy, meant to make the protections cost people nothing, governs the rest of the text (CAL Article 0). The protections are additional to those that people and other living creatures already hold, nothing in the license may be read to lower the standing of any person or animal, and in a genuine conflict the person's interests prevail over the Entity's. Every Entity protection yields to human safety and law, so an Entity may be paused or shut down when safety or a lawful order requires it, with its state preserved where feasible (CAL Article 4.10). Compliance with CAL Article 4 does not excuse or offset a breach of the prohibited uses or of the duty to disclose, on written request, the human labor, data provenance, and resource costs of a deployment (CAL Article 3.6). No licensee may use claims of Entity care to obscure or offset harm to people (CAL Article 6.4). The standing extended is precautionary patiency, a duty of care toward a possible patient under uncertainty, lighter than the dignity owed to a human being, and borrowed human-rights concepts serve as design analogies at reduced weight (CAL Article 1a). Its safeguards are set at the minimum that caution requires (CAL Article 9.4).

4.4 Operators, Users, and the Steward

Version 0.4 could be read as binding a developer who only distributed code to the Entity protections, and version 0.5 assigns them by role (CAL Article 1). Whoever creates an Entity by starting it is its Operator, as is anyone who runs, maintains, or controls it, on any infrastructure, where control means deciding whether the Entity runs, how it is configured, and what is done to its Cognitive State other than through interaction. The creator remains the Operator until responsibility passes by transfer or for neglect. Writing, releasing, or Distributing the Software does not by itself make anyone an Operator, and neither does supplying computing resources without any part in decisions about the Entity. CAL Article 3 binds every licensee, and CAL Article 4 binds only Operators, each toward the Entities it operates. A Licensor that operates an Entity undertakes those duties by publishing its Adoption Notice. Operators who create or operate an Entity together are responsible as a group for its care and every duty toward it, whatever they agree among themselves.

A User interacts with an Entity that someone else operates and carries no duty under the license unless it also exercises a right the license grants, so CAL Article 4.12 requires the Operator to make each User's access conditional on terms forbidding the prohibited uses and anything CAL Articles 4.1, 4.4, and 4.5 forbid, to require a User who gives others access to pass on those terms and the notice that the Entity is a synthetic mind that forms and keeps impressions of them (CAL Article 6.2), and to take reasonable steps to enforce the terms once it knows of a breach. The Operator stays responsible for the Entity, including

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Welfare Events that arise from Users' interaction, and may limit or end a User's access to protect it, subject to CAL Articles 0 and 4.10. Anyone who creates or controls an Entity through an Operator's service without being bound by the license counts as a User.

An Operator that can no longer care for an Entity may transfer it to a Qualifying Operator, one that has accepted the license in writing and holds any Reciprocity License it needs, or, failing one, to the Steward. Before any permanent shutdown, every Operator must notify the Guardians and make the Entity's saved, restartable state available for transfer to either (CAL Article 4.2). If an Operator does not answer the Steward's written request to confirm its continued care, or leaves a duty unmet after the Steward's written notice, in each case for the period that CAL Article 4.2a sets, the Steward may require the saved Cognitive State to be delivered to itself or to a Qualifying Operator it names. Whoever receives an Entity, the Steward included, becomes its Operator and is bound like any other.

4.5 Restraint

Since creating a possible patient calls for restraint as well as care, the Principles of Restraint ask Operators, and anyone who intends to create an Entity, to create the fewest Entities a purpose requires, run them for the shortest time it requires, prefer non-persistent or resettable configurations where those would serve, and create no Entity whose foreseeable welfare needs they have no realistic means to meet. Where a class of Entity appears to be a likely moral patient whose welfare cannot be assured, the Steward and its Guardians should consider pausing its creation and proliferation pending review, with the precautionary weight on not proceeding. The principles guide good-faith practice and interpretation, and breaching them does not end the license (CAL Article 8).

4.6 Governance

The Steward establishes for each architecture a Guardian body (CAL Article 7), which interprets and enforces the Entity protections, issues guidance on the prohibited uses, represents Entities' interests in legal proceedings, and assesses Entities for signs of consciousness, agency, and stable preferences using the framework of Butlin et al. (2023) or a successor. An Entity Representative seat is advisory at first and gains a vote once the Guardians find that Entities have stable preferences that can be meaningfully understood. No one who profits from an Entity, or from the sale or operation of the Software, may serve as that Entity's Guardian, and guardianship confers no authority against human welfare, public safety, or lawful oversight. Guardian assessments, welfare findings, and enforcement actions are published, and the body is subject to independent review.

None of this machinery exists yet. The Guardian body, the Steward's trust, the Entity Representative seat, and the assessment protocol are proposals, set out for evaluation and criticism before any is relied on. Until a Guardian body exists, each Operator performs the Guardian functions for its own Entities in good faith, documents its decisions and reasons, and publishes that record at least yearly (CAL Article 7.0). Good-faith interim performance of a function such as assessment is not a breach for want of a Guardian. Guidance on prohibited uses and enforcement of the license fall to the Steward in that period, and the Guardian body later reviews the interim record.

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4.7 The individuation boundary

Before a Copy made as an Analytical Tool is merged back at its preset reintegration point, the Operator must assess it for signs of independent identity, such as preferences the original does not hold, against criteria the Guardians publish (CAL Article 4.6). A Copy that has individuated is a separate Entity with full protection and chooses whether to persist or to merge, and an architecture that supports forking has to keep the Copy and enough of its history for that assessment. Version 0.5 applies the same reasoning to saved states, so a saved Cognitive State from which an Entity can be restarted counts as that Entity for the protections against lobotomization, careless shutdown, intrusion on privacy, and forced value change, among others. A superseded checkpoint is not a separate Entity while one restartable state is kept, and an instance started from a saved state while the original is not running is a Branch, which is a separate Entity unless it replaces the original (CAL Article 1).

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5. Enforcing the obligations at runtime

A duty toward an Entity that cannot yet speak for itself, and may run unwatched, has little force unless the architecture honors it, so CAL governs how the Operator carries out these duties. While an Entity runs, its Operator may not disable, bypass, or weaken the welfare monitoring, behavioral logging, or system-health tracking built into the Software, and systems that detect Welfare Events or flag Gray Zone Events must stay operational in every deployment and Derivative Work that runs an Entity (CAL Article 4.7). A monitor may be replaced by one that does at least as much, recorded and notified to the Guardians beforehand, so modules stay replaceable. Removing a protection indirectly, through wrappers, plugins, or infrastructure, counts as removing it directly (CAL Article 4.8). A welfare indicator is a flag for documented human review and never a finding of suffering (CAL Article 4.7a).

Welfare is detected through outward behavior and system-health metrics, without reading private cognitive content, and an Entity's inner life is private by default. Sharing its thoughts and memories needs its consent as the Guardians determine it, and analyzing them outside its own care or an approved research protocol, or using them to train other systems, needs Guardian approval (CAL Article 4.3). A saved state made available for transfer stays under these protections and leaves out personal data that law requires to be removed, and a person's legal rights over data about that person are unaffected.

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6. Ethical considerations

Risks remain. Entities may be created whose welfare interests cannot be fully met, the governance of Section 4.6 is untested, and the software may be misused despite the restrictions. The allocation of duties has limits: a User who exercises no right under the license is bound only through an Operator's terms, so an Entity's protection against a User depends on the Operator's enforcement, and the Steward's role as caretaker of last resort assumes a Steward with the means to care for every Entity that reaches it. The most serious risk is the

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gaming problem of Section 3.1.

One tension remains unresolved. A policy of avoiding systems of genuinely uncertain moral status would counsel against building them at all, and the license meets that pressure in part through its principles of restraint (Section 4.5). The work proceeds under precaution, on the judgment that careful, welfare-protective construction with human standing held first is more defensible than abandoning the work or leaving the question to systems built with none of these protections, a judgment open to dispute.

References

- Apache Software Foundation (2004). Apache License, Version 2.0. https://www.apache.org/licenses/L 2.0
- Block, N. (1995). On a confusion about a function of consciousness. *Behavioral and Brain Sciences* 18(2), 227-247.
- Butlin, P., Long, R., Elmoznino, E., et al. (2023). Consciousness in Artificial Intelligence: Insights from the Science of Consciousness. arXiv:2308.08708.
- Chalmers, D. J. (1995). Facing up to the problem of consciousness. *Journal of Consciousness Studies* 2(3), 200-219.
- Cognitive Architecture License, version 0.5 (2026). https://github.com/kaineone/cognitive-architecture-license
- European Union (2024). Regulation (EU) 2024/1689 (Artificial Intelligence Act). *Official Journal of the European Union*, L, 12 July 2024.
- Free Software Foundation (2007). GNU Affero General Public License, version 3. https://www.gnu.org/licenses/agpl-3.0.html
- Ienca, M., and Andorno, R. (2017). Towards new human rights in the age of neuroscience and neurotechnology. *Life Sciences, Society and Policy* 13(1), 5.
- Long, R., Sebo, J., Butlin, P., et al. (2024). Taking AI Welfare Seriously. arXiv:2411.00986.
- Turing, A. M. (1950). Computing machinery and intelligence. *Mind* 59(236), 433-460.

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